

SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF ECE
ECE 2022-2026 BATCH
PROJECT PHASE-1
ZEROth REVIEW

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Project Title:	High Performance SRAM Design for Energy Efficient Systems
Base Paper Title:	A Low-Power Variation-Tolerant 7T SRAM With Enhanced Read Sensing Margin for Voltage Scaling

Abstract:

Achieving high performance and energy efficiency in SRAMs is essential for modern voltage-scalable and low-power system-on-chip (SoC) applications. This work presents a robust SRAM architecture optimized to operate reliably under aggressive voltage scaling, while improving read sensing margin and reducing power consumption. The design incorporates techniques to suppress data-dependent bitline leakage and enhances variation tolerance without relying on external assist mechanisms. An adaptive timing generation scheme and an integrated error detection circuit ensure stable operation across process, voltage, and temperature (PVT) variations. Implemented as an 8-kb memory macro in 55 nm CMOS technology, the proposed design demonstrates error-free operation down to 0.35 V and achieves a minimum energy point of 1.76 pJ at 0.4 V. It also shows a significant reduction in leakage current and up to 19.3% average power savings compared to conventional SRAM implementations, making it a strong candidate for high-performance and energy-aware computing systems.

SIGNATURE OF GUIDE